MTH309 PRACTICE PROBLEMS WEEK 6

Question 1. Let f, g be two $\mathbb{R}$ valued simple functions on a measurable space. Prove that $f+g$ is again a simple function.

Question 2. Let $f = \sum_{i=1}^{m} a_i 1_{A_i} = \sum_{j=1}^{n} b_j 1_{B_j}$ be two representations of a non-negative simple function defined on a measure space $(\Omega, \mathcal{F}, \mu)$. Show that

$$\sum_{i=1}^{m} a_i \mu(A_i) = \sum_{j=1}^{n} b_j \mu(B_j),$$

i.e. the integral $\int_\Omega f \, d\mu$ is well-defined.

Question 3. Let $X : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathbb{B}_\mathbb{R})$ and $Y : (\Omega, \sigma(X)) \to (\mathbb{R}, \mathbb{B}_\mathbb{R})$ be Borel measurable. Show that there exists a Borel measurable function $f : \mathbb{R} \to \mathbb{R}$ such that $Y = f \circ X = f(X)$.
(Hint: Assuming $Y = 1_A$, try to find the explicit form of the function f.)

Question 4. Given an probability measure μ on $(\mathbb{R}, \mathbb{B}_\mathbb{R})$, consider the corresponding distribution function $F_\mu : \mathbb{R} \to [0,1]$ as

$$F_\mu(x) := \mu((-\infty, x]), \forall x \in \mathbb{R}.$$

Let μ and ν be two probability measures on $(\mathbb{R}, \mathbb{B}_\mathbb{R})$ with the same distribution function, i.e. $F_\mu(x) = F_\nu(x), \forall x \in \mathbb{R}$. Show that $\mu = \nu$, i.e. $\mu(A) = \nu(A), \forall A \in \mathbb{B}_\mathbb{R}$.

Question 5. Let $n \geq 2$ be a positive integer. Fix $x_1 < x_2 < \cdots < x_n$ in $\mathbb{R}$ and $p_1, p_2, \cdots, p_n \in [0,1]$ with $\sum_{i=1}^{n} p_i = 1$. Recall that $\mu = \sum_{i=1}^{n} p_i \delta_{x_i}$ is a probability measure on $(\mathbb{R}, \mathbb{B}_\mathbb{R})$. Compute the distribution function.

Question 6. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and fix $A \in \mathcal{F}$. Consider the random variable $X = 1_A$. Compute its law and distribution function.

Question 7. Let λ denote the Lebesgue measure on $(\mathbb{R}, \mathbb{B}_\mathbb{R})$. Show that

$$\lambda((a,b)) = \lambda((a,b]) = \lambda([a,b)) = \lambda([a,b]) = b - a$$

for all $-\infty < a < b < \infty$.

Question 8. Let λ denote the Lebesgue measure on $(\mathbb{R}, \mathbb{B}_\mathbb{R})$. For any fixed $x \in \mathbb{R}$, show that the set function $A \mapsto \lambda(A + x)$ is a measure. Here $A + x := \{a + x \mid a \in A\}$.

Question 9. Let λ denote the Lebesgue measure on $(\mathbb{R}, \mathbb{B}_{\mathbb{R}})$. For any fixed $x \in \mathbb{R}$, show that $\lambda(A) = \lambda(A + x), \forall A \in \mathbb{B}_{\mathbb{R}}$. This is usually stated as follows: The Lebesgue measure is translation invariant.

Question 10. Let $\prod_{j=1}^{d}(a_j, b_j]$ be a d-dimensional rectangle in $\mathbb{R}^d$. Observe that the co-ordinates of the vertices are made up of either a_j or b_j for each $j = 1, 2, \cdots, d$. Let Δ_k^d denote the set of vertices where exactly k many a_j's appear. Let $X = (X_1, X_2, \cdots, X_d)^t : (\Omega, \mathcal{F}, \mathbb{P}) \to (\mathbb{R}^d, \mathbb{B}_{\mathbb{R}^d})$ be a d-dimensional random vector. Show that

$$\mathbb{P}(X \in \prod_{j=1}^{d}(a_j, b_j]) = \mathbb{P}(a_1 < X_1 \leq b_1, a_2 < X_2 \leq b_2, \ldots, a_d < X_d \leq b_d) = \sum_{k=0}^{d}(-1)^k \sum_{x \in \Delta_k^d} F_X(x).$$